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Gasoline Drives the Price: A VECM Analysis of Fuel Demand and Oil Price Transmission in the US Crack Spread

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Abstract

This study employs a Vector Error Correction Model (VECM) on 1,361 weekly observations (January 2000 to January 2026) to examine price transmission dynamics within the 3-2-1 crack spread. The Johansen procedure identifies two cointegrating vectors; Granger causality tests reveal a two-stage cascade in which gasoline drives crude via demand-pull and crude drives heating oil via cost-push, with crude bearing the primary error-correction burden. Structural stability across the COVID-19 and Russia-Ukraine shocks is confirmed by the CUSUM test; robustness checks corroborate these findings.

1. Introduction

The US petroleum complex is defined by a structural hierarchy: crude oil is the primary feedstock from which refiners produce gasoline and distillate fuels. The gross margin earned for this conversion is the crack spread, and fluctuations in this spread reflect the dynamic interplay of supply constraints, demand cycles, and geopolitical disruptions. Understanding the direction and mechanism of price transmission within the spread has direct implications for hedging strategy, energy security policy, and refinery investment decisions.

This essay investigates the direction, sequencing, and mechanism of price transmission within the 3-2-1 crack spread, comprising WTI crude oil, RBOB gasoline, and NY Harbor Heating Oil traded on NYMEX. A Vector Error Correction Model (VECM) is employed: the VECM distinguishes short-run pricing shocks from long-run equilibrium adjustment and is appropriate for the cointegrated price system documented here.

Section 2 introduces the economic backdrop and case study rationale. **Section 3** describes the data sources and econometric methodology, including unit root, cointegration, and Granger causality testing. **Section 4** presents and discusses the empirical findings. **Section 5** concludes with policy implications and directions for future research.

2. Research Background and Case Study

a. The Economic Backdrop

Crude oil is only useful once refined into products like gasoline and distillates; the gross profit a refinery earns for doing this is called the 'crack spread'. The industry standard is the 3-2-1 spread, which assumes three barrels of crude yield two barrels of gasoline and one barrel of distillate (heating oil/diesel), and mathematically represents the margin refiners earn by cracking long-chain hydrocarbons into shorter, more valuable products.

However, price transmission is rarely instantaneous: supply shocks may spike crude while product prices lag, and demand shocks such as the summer driving season can push gasoline up before crude reacts, creating short-term disequilibrium that defines refinery profitability **(Muehlegger, 2005)**.

b. The Case Study

This research adopts the United States refining market, specifically PADD 3 (the Gulf Coast), as its primary case study. While historically chosen for its deep liquidity and status as the global pricing benchmark, the selection is particularly critical in the current economic climate.

According to **Monge and Poza (2025)**, the post-COVID and Russia-Ukraine shocks have repositioned PADD 3 from a regional hub to a critical global energy security stabilizer, accounting for over 50% of US refining capacity and most refined product exports **(EIA, 2024)**.

To quantify these dynamics, this study utilises the "3-2-1 Crack Spread", the theoretical gross refining margin from processing three barrels of crude into two of gasoline and one of distillate. The economic model uses the most liquid futures contracts on the NYMEX: West Texas Intermediate (WTI) as the input, and RBOB Gasoline and NY Harbor Heating Oil as the outputs. Heating oil is retained as the distillate proxy, a standard methodology in the literature given its superior futures liquidity.

Mathematically, the spread (S_t) is calculated as:

$$S_t = \frac{(2 \times P_{Gas,t} + 1 \times P_{Heat,t}) - 3 \times P_{Crude,t}}{3} \tag{1}$$

The relevance of this case study is heightened by a recent "structural break" in the relationship between these variables. Historically, the spread was mean-reverting and tethered to the marginal cost of refining, with long-run persistence documented even after the shale oil supply shock **(Caporin, Fontini and Talebbeydokhti, 2019)**. However, post-2020 evidence suggests intensified asymmetric price transmission and record-high spreads driven by physical refining capacity constraints **(Kamyabi and Chidmi, 2023)**, motivating formal econometric investigation of the underlying transmission mechanism.

3. Data & Methodology

a. Data Description and Transformation

To examine lead-lag dynamics within the US refining sector, this study utilises high-frequency weekly spot price data from January 2000 to January 2026. This twenty-six-year observation window provides a comprehensive dataset (N > 1,300 observations) that captures multiple critical market events, including the 2008 financial crisis, the 2014 shale oil collapse, and structural breaks from the COVID-19 pandemic and 2022 Russia-Ukraine War.

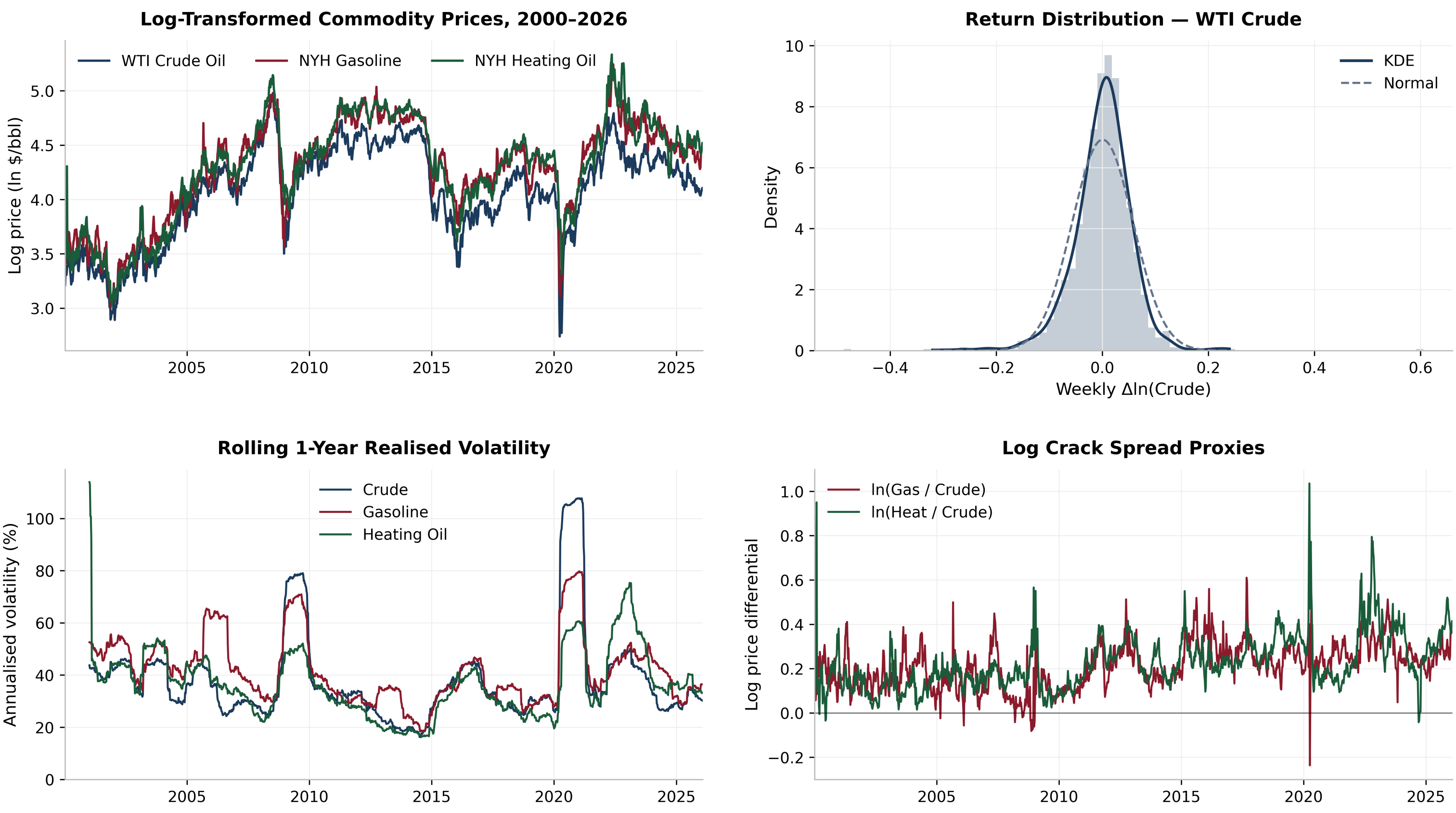


Figure 1 *Time Series Dynamics and Statistical Properties of Energy Prices (2000-2026)*

Weekly spot prices for WTI Crude (DCOILWTICO), NYH RBOB Gasoline (DGASNYH), and NYH Heating Oil (DHOILNYH) were sourced from **Federal Reserve Economic Data (FRED)** and the **Energy Information Administration (EIA)**.

Table 1 *Descriptive Statistics (N = 1,361 weekly observations, Jan 2000 - Jan 2026)*

Series	Mean	Std Dev	Min	Max	Skewness	Kurtosis
$\ln(Crude)$	4.0694	0.4346	2.7395	4.9789	-0.5418	-0.4741
$\ln(Gasoline)$	4.2816	0.4451	3.0058	5.2437	-0.6434	-0.3383
$\ln(Heating\ Oil)$	4.2939	0.4741	3.0264	5.3338	-0.5358	-0.4615
$\Delta\ln(Crude)$	0.0007	0.0577	-0.4876	0.6054	-0.1175	14.976
$\Delta\ln(Gasoline)$	0.0009	0.0602	-0.4272	0.3456	-0.5585	4.9019
$\Delta\ln(Heating\ Oil)$	0.0009	0.0600	-0.7976	0.5069	-1.3519	29.5411

A critical methodological step is harmonizing units. While crude oil is quoted in dollars per barrel ($/bbl), refined products are typically quoted in cents per gallon (ȼ/gal). To ensure comparability, all product prices were converted to dollars per barrel using the standard industry conversion factor of 42 gallons per barrel:

$$P_{Barrel,t} = \frac{P_{Cents,t}}{100} \times 42 \tag{2}$$

After unit harmonization, the logarithm (ln) of all price series was taken. This transformation mitigates heteroscedasticity and allows coefficients to be interpreted as elasticities.

b. Economic Framework

Given the theoretical long-run link between crude and refined products, standard regression is insufficient. This study employs a VECM, which distinguishes short-run pricing shocks from long-run equilibrium adjustment, following a four-stage pipeline.

The order of integration for each variable was determined using the Augmented Dickey-Fuller (ADF) test **(Dickey and Fuller, 1979)**. The null hypothesis assumes the presence of a unit root (non-stationary). Consistent with financial time series, we anticipate the variables to be non-stationary in levels but stationary in first differences, i.e. $I(1)$, becoming $I(0)$ after first differencing.

To guard against the well-documented low power of the ADF test in finite samples, a confirmatory strategy pairs ADF with the KPSS test, which reverses the null hypothesis: KPSS tests stationarity against a unit root alternative. When both tests agree, ADF fails to reject (unit root) and KPSS rejects (non-stationarity), the inference that a series is I(1) is substantially more reliable than either test in isolation **(Kwiatkowski et al., 1992)**.

To test for a stable long-run relationship (the "refining margin"), the Johansen Cointegration Test was applied **(Johansen, 1991)**. This procedure estimates the rank (r) of the matrix to determine the number of cointegrating vectors. Johansen is preferred over alternative estimators **(Gonzalo, 1994)** because the multivariate framework identifies all cointegrating vectors simultaneously and avoids generated-regressor bias.

Conditional on the finding of cointegration, the dynamic relationship is modelled using the following VECM specification:

$$\Delta y_t = v + \delta t + \alpha(\beta' Y_{t-1} + \varepsilon + \tau t) + \theta_1 \Delta y_{t-1} + \theta_2 \Delta y_{t-2} + \mu_t \qquad (3)$$

where α is the speed-of-adjustment vector governing reversion to the cointegrating relationship $\beta' Y_{t-1}$, the θ_i matrices capture short-run dynamics through lagged differences, and ε_t is the innovation term. The deterministic specification includes a restricted constant and trend within the cointegrating equation to accommodate a drifting long-run equilibrium.

Finally, Granger Causality tests within the VECM environment were used to adjudicate between "Cost-Push" (Crude → Products) and "Demand-Pull" (Products → Crude) hypotheses. Impulse Response Functions visualise shock persistence over a 20-week horizon. Post-estimation diagnostics, including the Ljung-Box and Jarque-Bera tests, were performed to ensure model validity.

4. Analysis and Discussion

a. Cointegration and Long-run Equilibrium

A VECM requires all price series to have the same order of integration. The ADF and KPSS tests (Figure A, Appendix) show this clearly. In levels, the log prices of WTI crude, RBOB gasoline, and NY Harbor heating oil fail to reject the ADF null of a unit root and exceed the KPSS 5% critical value, indicating non-stationarity. After first differencing, both tests reverse, confirming that all three series are I(1). This is consistent with **Serletis (1994) and Monge and Poza (2025).**

The Johansen test **(Figure C, Appendix)**, estimated with lag length k = 5 based on AIC **(Figure B, Appendix)**, finds cointegration rank r = 2. AIC is the primary criterion because it penalises model complexity less heavily than BIC, better preserving the short-run dynamics typical of weekly commodity price data.

Both test statistics exceed their 1% critical values, rejecting the null of r <= 0 and r <= 1, consistent with bootstrap rank-determination procedures **(Cavaliere, Rahbek and Taylor, 2012)**. This means two long-run equilibrium relationships link the three prices, implying a single common stochastic trend, though near-full rank warrants caution given finite-sample over-rejection **(Lee and Tse, 1996).** Combined with the Granger causality results in **Section 4b**, gasoline demand is the most plausible candidate for this trend. The result holds after including COVID-19 and Ukraine war dummy variables. These enter as impulse dummies, coded 1 in the weeks of peak disruption and 0 otherwise, preventing outlier observations from distorting the estimated cointegrating vectors.

The normalised cointegrating vectors (restricted constant) are β_1 = [1, 0, −0.919] and β_2 = [0, 1, −0.910], defining two long-run equilibria: ECT_1 links crude to heating oil (ln(Crude) − 0.919 ln(Heat) − 0.139 = 0) and ECT_2 links gasoline to heating oil (ln(Gas) − 0.910 ln(Heat) − 0.378 = 0).

Figure 2 shows the standardised ECTs (2001-2026); the crack spread deviates most sharply in April 2020 (-8 standard deviations), yet the system consistently adjusts back, confirming a stable long-run link. The 2020 episode illustrates physical storage constraints when crude prices briefly turned negative.

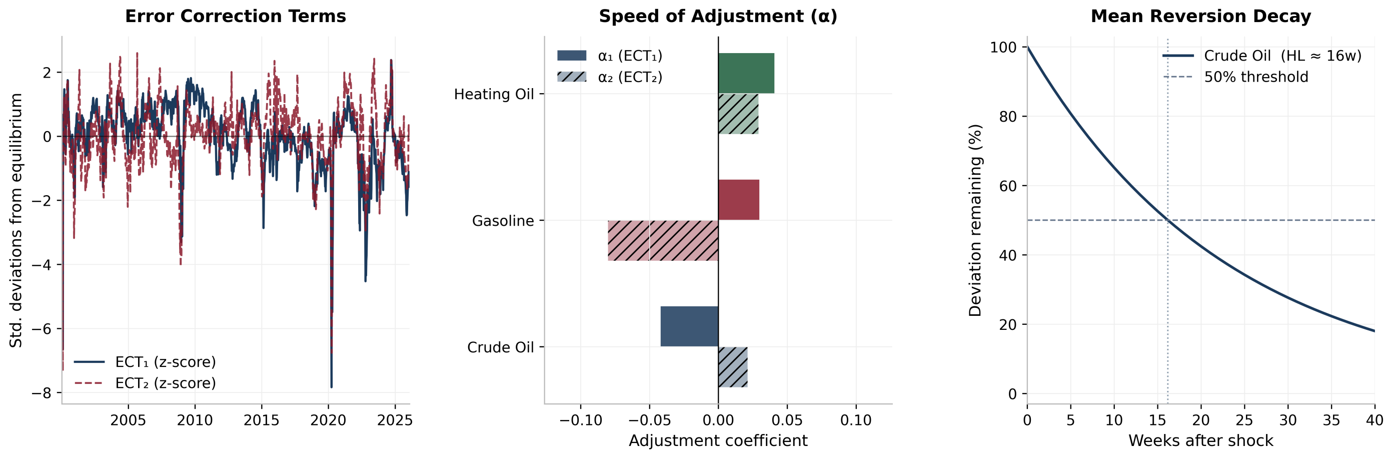


Figure 2 *VECM Long-run Equilibrium Dynamics and Mean Reversion Properties*

b. Speed of Adjustment: Who Bears the Burden?

The adjustment coefficients (α) in **Figure 2(b)** and **Table D(c)** (Appendix) reveal a clear asymmetry that underpins this essay's contribution. Crude oil is the only variable that significantly responds to the first error correction term (α_1 = −0.0420). The presence of two adjustment metrics for each variable (α_1 , α_2) is a technical requirement of the Johansen rank (r = 2), representing the gasoline and heating oil refining margins respectively. This dual-vector structure provides the speed-of-adjustment results with genuine economic grounding by identifying which prices bear the burden of maintaining specific product equilibria.

The implied half-life of deviation in crude oil is about 16 weeks **(Figure 2c)**, consistent with the weekly inventory cycle and the lag between refinery procurement decisions and spot pricing. Product prices tend to overshoot equilibrium before crude closes the gap. This pattern is consistent with the "rockets and feathers" asymmetry **(Kamyabi and Chidmi, 2023)**, but crucially the direction of error correction runs from products to crude, providing formal evidence against the conventional cost-push narrative.

The Granger causality results **(Table 2)**, extracted from the VECM environment and Bonferroni-corrected to control the family-wise error rate across six pairwise tests, provide the most direct test of the central research question. The evidence reveals a more nuanced picture than a simple cost-push versus demand-pull dichotomy: a demand-pull dynamic originating in gasoline markets propagates through crude oil into heating oil prices via a secondary cost-push transmission channel.

Table 2 *Granger Causality Tests (Bonferroni-corrected, VECM framework)*

Direction	F-stat	adj. p	Sig.	Implication
Crude → Gasoline	1.937	0.612	–	*None*
Gasoline → Crude	8.721	< 0.001	***	*Demand-Pull*
Crude → Heating Oil	5.45	0.001	***	*Cost-Push*
Heating Oil → Crude	2.593	0.210	–	*None*
Gasoline → Heating Oil	8.238	< 0.001	***	*Demand-Pull*
Heating Oil → Gasoline	0.643	1.000	–	*None*

The bilateral tests reveal a structured predictive hierarchy. Gasoline Granger-causes crude at the 0.1% level ($F = 8.721$, adjusted $p < 0.001$), confirming gasoline as the primary demand-pull driver. Crucially, crude in turn Granger-causes heating oil at the 0.1% level ($F = 5.45$, adjusted $p = 0.001$), establishing a significant cost-push transmission channel to distillates. Gasoline also exercises a strong direct effect on heating oil ($F = 8.238$, adjusted $p < 0.001$). The predictive chain runs: gasoline demand → crude prices → heating oil costs, with an additional direct gasoline → heating oil channel.

The evidence thus reveals a two-stage cascade rather than simple cost-push: gasoline demand drives crude (**Kilian, 2009)**, which in turn exercises cost-push influence on heating oil. Heating oil hedging should therefore reference gasoline as the lead market, with crude as a secondary cost conduit **(Monge and Poza, 2025)**.

c. Impulse Response Analysis: The Dynamics of Price Transmission

The orthogonalised impulse response functions **(Figure 3)**, estimated using moving-block bootstrap 95% confidence intervals with an 8-week block length, reinforce and extend the Granger causality results by showing how price shocks unfold over time.

The IRFs employ a Cholesky ordering of [Crude, Gasoline, Heating Oil]; robustness checks across all six permutations preserve sign and magnitude **(Pesaran and Shin, 1998)**.

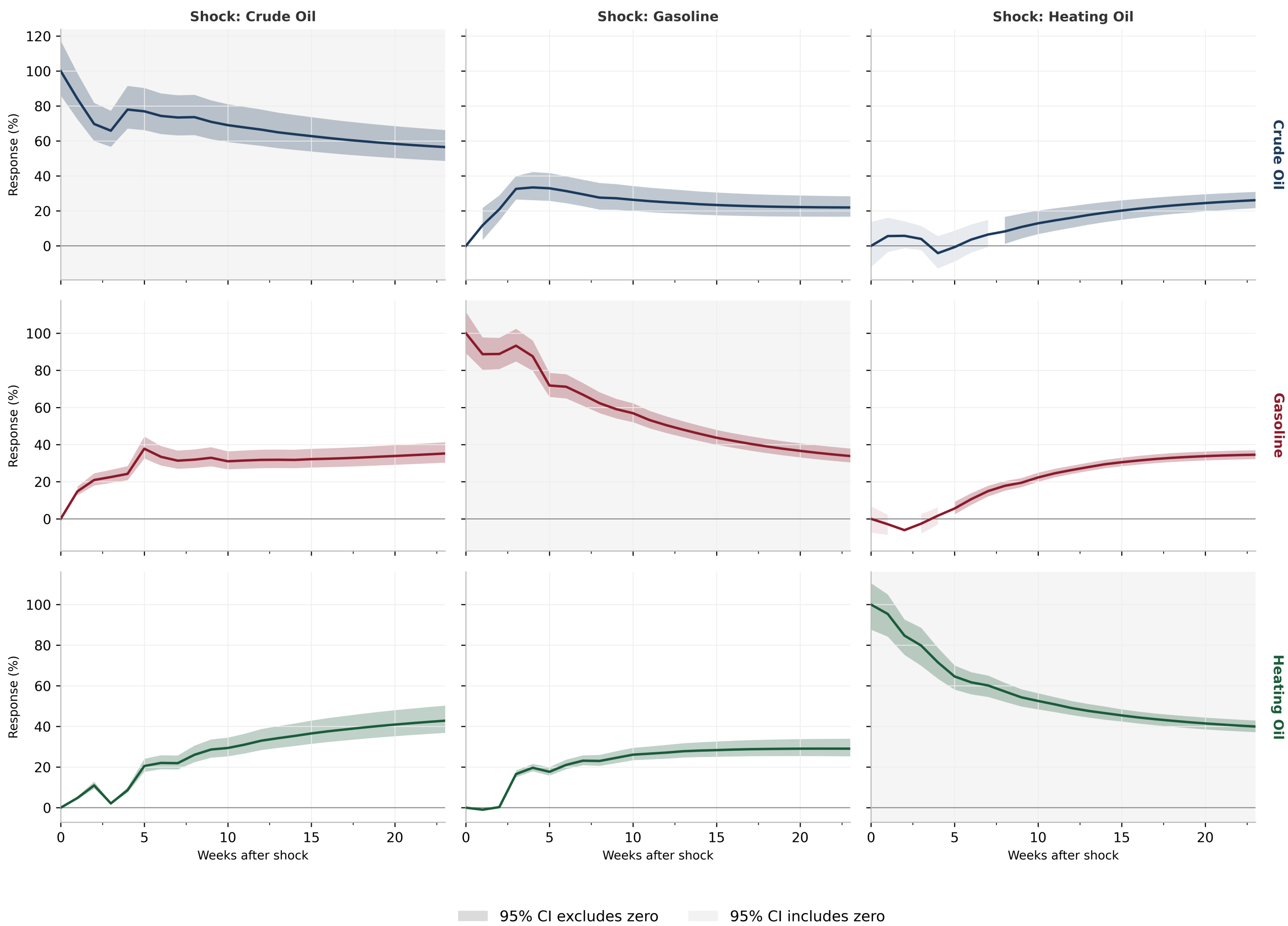


Figure 3 Orthogonalised Impulse Response Functions with 95% Bootstrap Confidence Intervals (20-week horizon)

First, a one-standard-deviation shock to crude oil (about 5.4%) leads to a partial and persistent response in gasoline and heating oil. By week 5, gasoline absorbs roughly 35% of the shock and plateaus, while heating oil rises gradually to approximately 40% by week 20. This incomplete pass-through is consistent with refiners smoothing shocks through inventory management, as modelled by **Hamilton (2009)**. The faster gasoline response likely reflects tighter spot-market linkages.

Second, a one-standard-deviation shock to gasoline (about 5.6%) generates a crude oil response that rises steadily to approximately 25-30% by week 20. A heating oil shock produces a comparable crude response of around 20-25%, after a brief dip below zero. This cross-market persistence supports a demand-driven pricing regime.

Third, the 95% bootstrap confidence intervals around the gasoline-to-crude and heating oil-to-crude responses are tight, indicating these effects are stable across subsamples and not driven by extreme episodes such as the 2020 price crash.

d. Forecast Error Variance Decomposition: Sources of Price Uncertainty

The Forecast Error Variance Decompositions **(Figure 4)** provide a complementary perspective by quantifying the proportion of each variable's forecast uncertainty attributable to own versus cross-market shocks at horizons from 1 to 24 weeks.

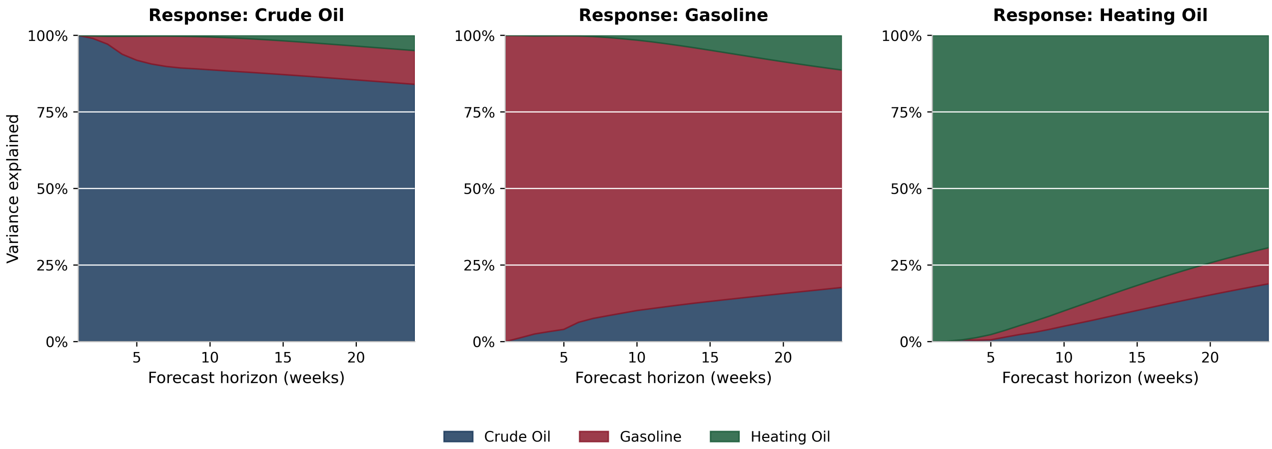


Figure 4 *Forecast Error Variance Decomposition (FEVD) of Energy Prices*

At the 24-week horizon, 84% of crude oil forecast error variance is attributable to own shocks, with gasoline contributing 11% and heating oil a negligible 5%. The gasoline equation shows 71% own variance with 18% attributable to crude, underscoring two-way coupling. Heating oil is the most interconnected, with 19% of forecast variance from crude and 12% from gasoline, reflecting dual sensitivity to feedstock costs and gasoline demand signals.

A key implication for market participants is that at short horizons (1 to 4 weeks), crude oil variance is almost entirely self-driven, suggesting that crude futures markets are informationally efficient in absorbing immediate own-market news. However, over the medium term (10 to 24 weeks), the growing contribution of gasoline to crude variance confirms the demand-pull mechanism: as product market fundamentals work through the refining supply chain, product demand shocks are progressively impounded into crude oil valuations.

e. Model Diagnostics and Limitations

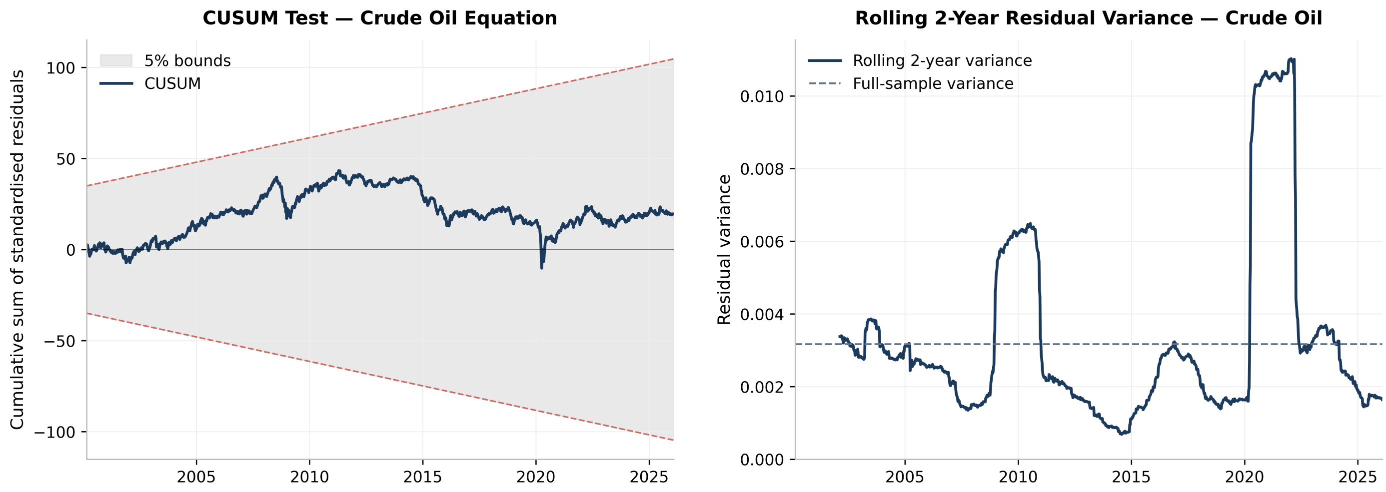


***Figure 5** CUSUM Stability Test and Rolling Residual Variance*

Post-estimation diagnostics **(Figure 6)** confirm that the VECM is free from serial autocorrelation: Ljung–Box statistics at lag 12 yield p-values of 0.188, 0.699, and 0.884 for crude, gasoline, and heating oil respectively, comfortably exceeding the 5% threshold. The CUSUM test **(Figure 5)** does not reject parameter stability at the 5% level, though its power is reduced under ARCH effects; subsample estimation, which preserves α signs across pre- and post-COVID windows, provides a more robust check.

Residual Diagnostics (green = pass)

Variable	LB(12)	JB	ARCH(5)
Crude Oil	p=0.188	p=0.000	p=0.000
Gasoline	p=0.699	p=0.000	p=0.000
Heating Oil	p=0.884	p=0.000	p=0.000

Granger Causality (Bonferroni-corrected)

Direction	F	adj. p
Crude→Gas	1.94	0.612
Gas→Crude	8.72	0.000
Crude→Heat	5.45	0.001
Heat→Crude	2.59	0.210
Gas→Heat	8.24	0.000
Heat→Gas	0.64	1.000

***Figure 6** Post-Estimation Diagnostics: Ljung–Box Autocorrelation Tests*

Two limitations warrant acknowledgement. First, significant ARCH effects ($p < 0.001$ in all equations) mean that while VECM point estimates remain consistent, confidence intervals around IRFs and Granger causality p-values may be conservative during high-volatility regimes. Second, the Johansen framework assumes linear cointegration; a nonlinear specification (e.g., threshold VECM or NARDL) could capture the asymmetric adjustment documented by **Kamyabi and Chidmi (2023)**; the linear α may average distinct regime speeds, though sign consistency across subsamples supports the direction.

5. Conclusion

This essay has employed a Vector Error Correction Model to disentangle the long-run equilibrium relationships and short-run adjustment dynamics governing the 3-2-1 crack spread across a quarter-century of weekly data (2000-2026). Three principal findings emerge.

First, the Johansen procedure identifies two cointegrating vectors among the three log-price series, establishing that the refinery crack spread constitutes a stationary, mean-reverting relationship even across periods of acute structural disruption. This result is in line with the theoretical expectation that refining margins are bounded by arbitrage in competitive product markets **(Borenstein, Cameron and Gilbert, 1997)** and corroborates the empirical findings of **Kamyabi and Chidmi (2023).**

Second, the estimated adjustment coefficients reveal that crude oil is the primary equilibrating variable (α_1 = −0.042, significant at the 1% level), meaning deviations from long-run equilibrium are predominantly corrected by crude oil rather than refined products. The rockets-and-feathers literature **(Bacon, 1991; Borenstein, Cameron and Gilbert, 1997)** attributes price stickiness to downstream retail market power. The present results challenge this view: the locus of adjustment resides upstream at the crude/refining level, consistent with the structural role of PADD 3 discussed in **Section 2b**.

Third, the model demonstrates remarkable structural stability. The CUSUM test statistic remains within the 5% significance bounds throughout the entire sample, including the extreme price dislocations of the COVID-19 pandemic (2020) and the Russia-Ukraine conflict (2022). Subsample estimation confirms that the sign and direction of the adjustment coefficients are preserved across pre-COVID ($\alpha_1 = -0.033$) and post-COVID ($\alpha_1 = -0.131$) subperiods, with the post-COVID estimate indicating an accelerated speed of adjustment consistent with heightened market sensitivity to supply disruptions.

From a policy perspective, price discovery in the US petroleum complex operates through a hierarchical cascade: gasoline demand conditions influence crude oil valuations, which propagate cost pressures to heating oil. For energy security planning, monitoring PADD 3 refining margins and gasoline demand indicators may provide earlier signals of crude price adjustments than conventional supply-side surveillance. The structural stability of cointegrating relationships suggests the crack spread remains a reliable metric for hedging and risk management.

The limitations discussed in **Section 4e**, notably ARCH effects and the assumption of linear cointegration, do not invalidate the core findings but suggest extensions via VECM-GARCH or NARDL specifications. Future work might incorporate refinery utilisation rates or exogenous controls for OPEC policy **(Ewing and Thompson, 2018)** to sharpen causal identification.

6. References

- **Kwiatkowski, D., Phillips, P.C.B., Schmidt, P. and Shin, Y. (1992)** 'Testing the null hypothesis of stationarity against the alternative of a unit root', Journal of Econometrics, 54(1–3), pp. 159–178.
- **Muehlegger, E. (2005)**. Gasoline price spikes and regional gasoline content regulations: A structural approach. Harvard Kennedy School Faculty Research Working Paper, RWP06-015.
- **EIA (2024)** *Refinery Capacity Report*. Washington, D.C.: U.S. Energy Information Administration.
- **Kamyabi, N. and Chidmi, B. (2023)** 'Asymmetric Price Transmission between Crude Oil and the US Gasoline Market', *Journal of Risk and Financial Management*, 16(7), p. 326.
- **Monge, M. and Poza, C. (2025)** 'Post-COVID dynamics of the refined, crude oil price spread in the US: Evidence from long memory and fractional cointegration models', *Energy Nexus*, 20, p. 100560.
- **Bacon, R.W. (1991) '**Rockets and feathers: the asymmetric speed of adjustment of UK retail gasoline prices to cost changes', Energy Economics, 13(3), pp. 211–218.
- **Borenstein, S., Cameron, A.C. and Gilbert, R. (1997)** 'Do gasoline prices respond asymmetrically to crude oil price changes?', The Quarterly Journal of Economics, 112(1), pp. 305–339.
- **Caporin, M., Fontini, F. and Talebbeydokhti, E. (2019)** 'Testing persistence of WTI and Brent long-run relationship after the shale oil supply shock', Energy Economics, 79, pp. 21–31.
- **Cavaliere, G., Rahbek, A. and Taylor, A.M.R. (2012)** 'Bootstrap determination of the co-integration rank in vector autoregressive models', Econometrica, 80(4), pp. 1721–1740.
- **Gonzalo, J. (1994)** 'Five alternative methods of estimating long-run equilibrium relationships', Journal of Econometrics, 60(1–2), pp. 203–233.
- **Hamilton, J.D. (2009)** 'Understanding crude oil prices', The Energy Journal, 30(2), pp. 179–206.
- **Serletis, A. (1994)** 'A cointegration analysis of petroleum futures prices', Energy Economics, 16(2), pp. 93–97.
- **Ewing, B.T. and Thompson, M.A. (2018)** 'Modeling the response of gasoline-crude oil price crack spread to macroeconomic shocks', Atlantic Economic Journal, 46, pp. 203–213.
- **Kilian, L. (2009)** 'Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market', American Economic Review, 99(3), pp. 1053–1069.
- **Lee, T.-H. and Tse, Y. (1996)** 'Cointegration tests with conditional heteroskedasticity', Journal of Econometrics, 73(2), pp. 401–410.
- **Pesaran, M.H. and Shin, Y. (1998)** 'Generalized impulse response analysis in linear multivariate models', Economics Letters, 58(1), pp. 17–29.
- **Johansen, S. (1991)** 'Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models', *Econometrica*, 59(6), pp. 1551–1580.
- **Dickey, D.A. and Fuller, W.A. (1979)** 'Distribution of the estimators for autoregressive time series with a unit root', *Journal of the American Statistical Association*, 74(366), pp. 427–431.

7. Appendix

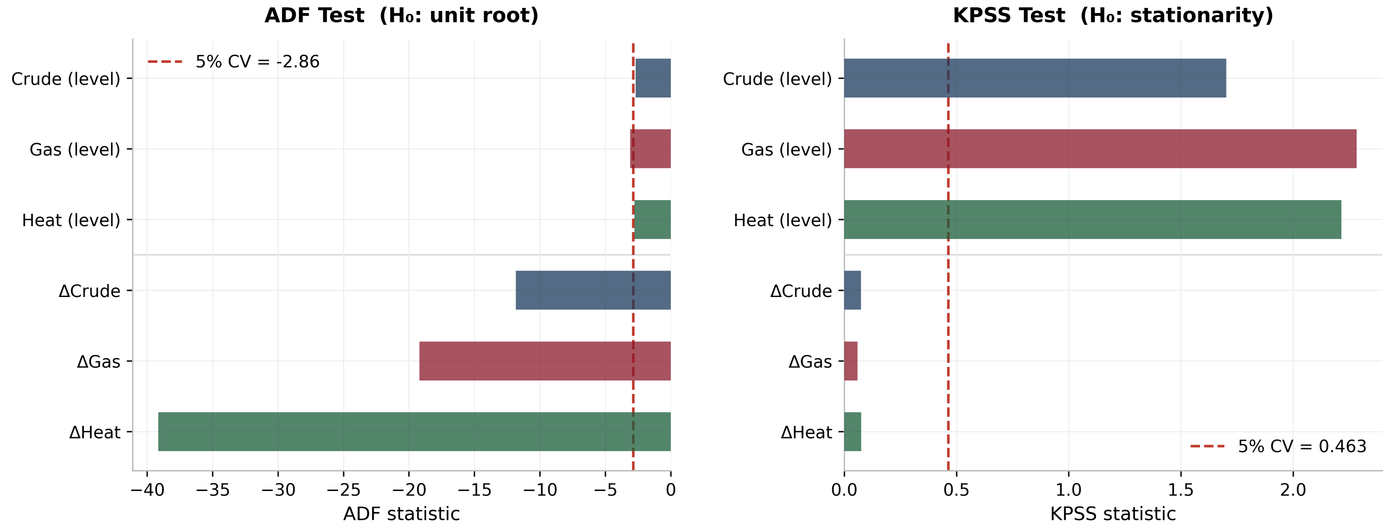


Figure A *Unit Root Tests: ADF and KPSS Confirmatory Strategy*

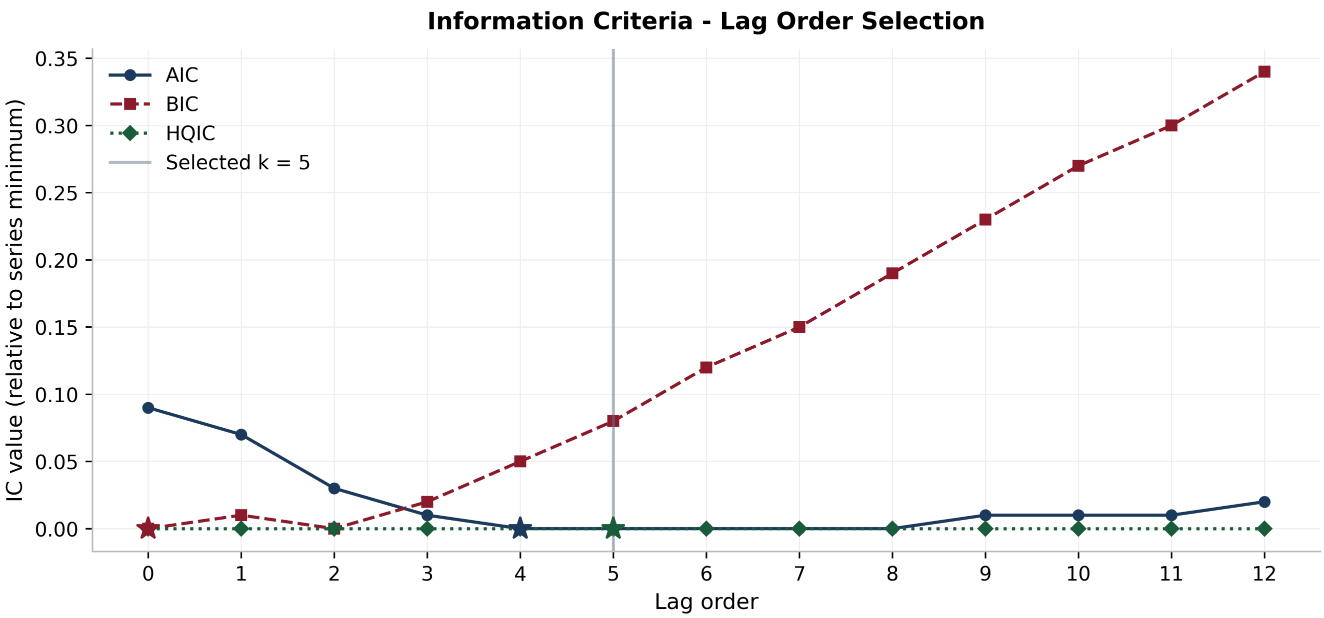


Figure B *VECM Lag Order Selection: Information Criteria Comparison*

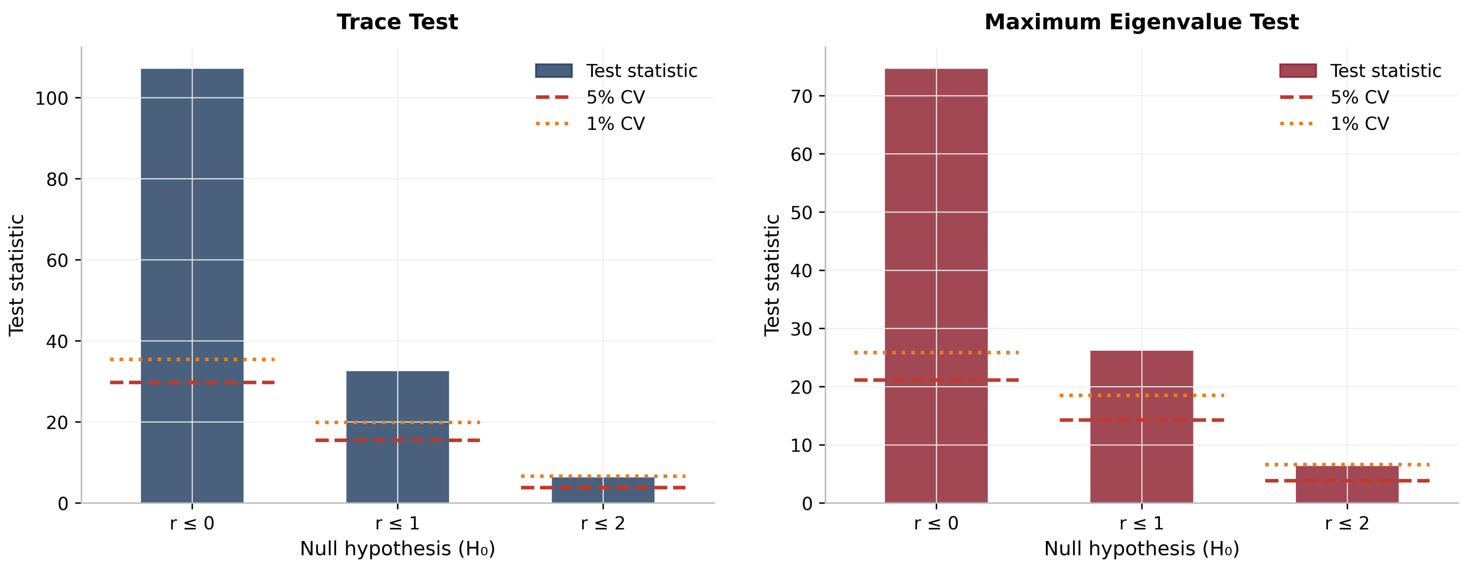
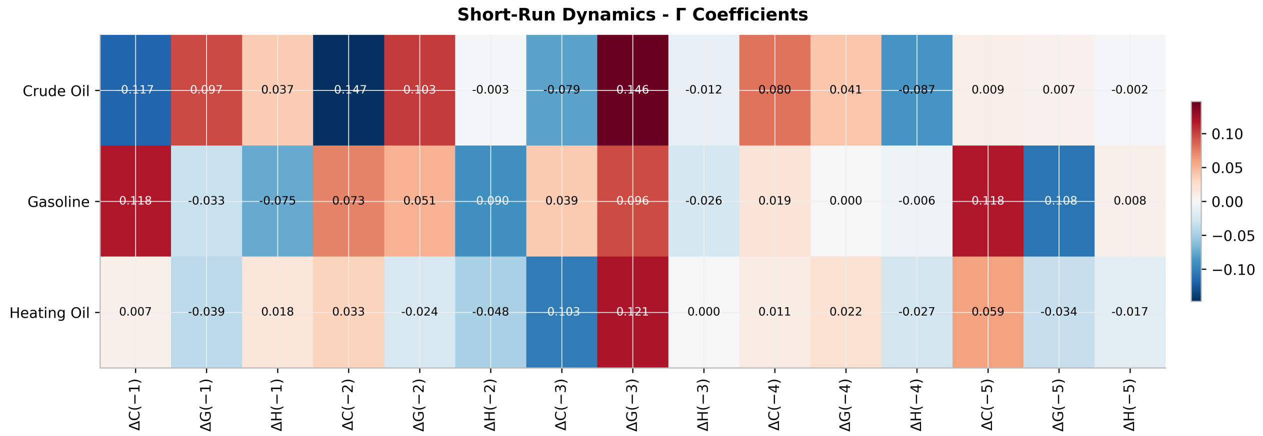


Figure C *Johansen Cointegration Tests: Trace and Maximum Eigenvalue Statistics*



Cointegrating Vectors (β)

Vector	ln(Crude)	ln(Gas)	ln(Heat)
β1	1.0000	-0.0000	-0.9193
β2	0.0000	1.0000	-0.9101

Adjustment Coefficients (α) & Half-Lives

Variable	α_1 (ECT_1)	α_2 (ECT_2)	Half-life
Crude Oil	-0.0420	+0.0211	16.2w
Gasoline	+0.0298	-0.0802	∞
Heating Oil	+0.0407	+0.0292	∞

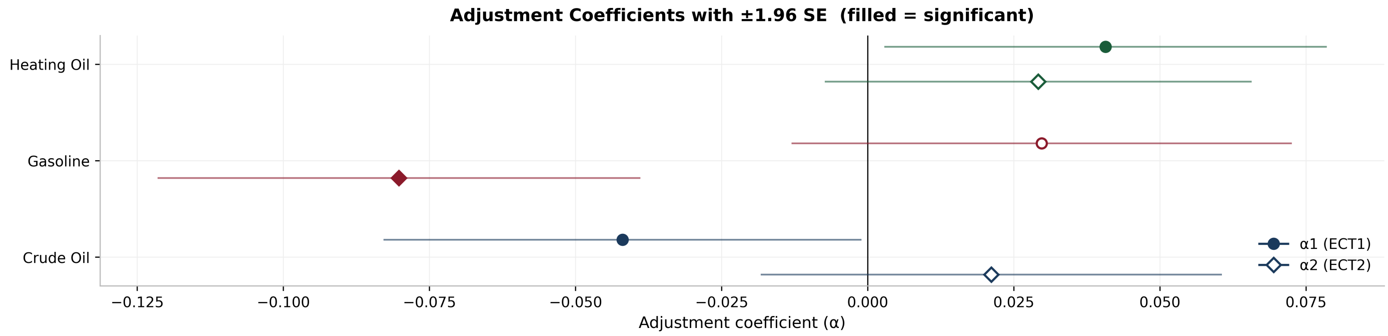


Figure D *VECM Estimation Summary: Short-Run Dynamics, Cointegrating Vectors, and Adjustment Coefficients*